

RoyaleCoach

Software Developer & Computer Science Student

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SUMMARY

I'm an undergraduate student majoring in Informatics Engineering at Institut Teknologi Sepuluh Nopember (ITS), under the Faculty of Intelligent Electrical and Informatics Technology (FTEIC). I've been writing code for a few years now, and I still find it fascinating how a few lines can turn an idea into something real.

TECHNICAL SKILLS

Languages: Python, Go, JavaScript

Systems: Rust, C#, C/C++

Engineering: Architecture, Algorithms, OOP

Domains: Cryptography, AI, Linux

Tools: Linux, Security First, Polyglot, Linux User, Clean Architect, Self-Taught

OS: Linux (Debian)

PROJECTS

Cryptid (Python) — Production-grade CLI cryptographic toolkit with AES-256-GCM and Argon2id. Secure by default, cross-platform.

Technologies: Python 3.13+, AES-256-GCM, Argon2id, Typer, Rich

- AES-256-GCM authenticated encryption
- Argon2id key derivation (256MB, 3 iterations)
- File encryption with metadata headers

RPG-Game (Python) — Terminal-based RPG with turn-based combat, skill trees, inventory system, equipment rarities, and SQLite persistence.

Technologies: Python, SQLite, OOP, CLI

- Turn-based combat with shield mechanic
- Skill trees for character progression
- Equipment rarities (Common to Mythic)

QRGen (Python) — Terminal QR code generator with Unicode block rendering, ASCII fallback mode, and automatic terminal size validation.

Technologies: Python, CLI, Unicode, QR Code

- Unicode block character rendering
- ASCII fallback for any terminal
- Terminal size validation

NoteApp (Python) — Desktop note-taking application with clean architecture, search functionality, and organized note management.

Technologies: Python, OOP, GUI, JSON

- Create & edit notes with auto-save
- Full-text search with keyword highlighting
- Categories & tags with color coding

EDUCATION

Institut Teknologi Sepuluh Nopember (ITS)

Informatics Engineering — FTEIC

Currently pursuing undergraduate degree

EXPERIENCE & JOURNEY

Now — Studying at ITS — Building Real Things

Currently pursuing Informatics Engineering at ITS under FTEIC. University gave me the theoretical foundation — algorithms, data structures, operating systems — but the real growth happens when I apply those concepts to actual projects.

2026 — Cryptography, AI & System Design

Built Cryptid, a CLI cryptographic toolkit using AES-256-GCM and Argon2id — my first serious attempt at security-first design. Explored AI agent architecture by building a hierarchical memory system with reflection engines.

2025 — Building Projects & Finding My Stack

Created RPG-Game — a terminal RPG that taught me more about OOP, debugging, and code review than any tutorial could. Built QRGGen and NoteApp to solve real problems I had. Switched to Linux as my daily driver.

2024 — Learning Beyond Python

After getting comfortable with Python, I started exploring other languages — Go for its simplicity and concurrency, Rust for its safety guarantees, C# for its type system, and JavaScript for the web.

KEY ACHIEVEMENTS

- **ITS Student** — Informatics Engineering at Institut Teknologi Sepuluh Nopember
- **Security First** — Built a production-grade cryptographic toolkit from scratch
- **Polyglot** — Proficient in 6+ programming languages
- **Linux User** — Debian daily driver — terminal-first workflow
- **Clean Architect** — Practices SOLID, Clean Architecture, and DRY principles
- **Self-Taught** — Started coding during the pandemic — now building real projects

LANGUAGES

- Indonesian (Native)
- English (Professional)

CURRENTLY EXPLORING

Rust, AI Agents, Cryptography, System Design